

# Financial System Integration and Shock Transmission

1MT404 Financial Stability

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# Agenda

- 1 The Changing Financial Landscape
- 2 Network and Linkages: Banks vs. NBFIs
- 3 The Rise of Private Credit
- 4 Channels of Shock Transmission
- 5 Historical Episodes & Lessons
- 6 Vulnerabilities & Policy Implications
- 7 Conclusion

# The Great Migration: From Banks to Non-Banks

- **Historical Pivot:** Since the Global Financial Crisis (2008), the financial system has undergone a "de-banking" process. While banks were constrained by strict regulation (Basel III), financial activity migrated to the less regulated NBFIs sector.
- **Total Assets:** Global financial assets reached approximately **\$500 trillion** in 2024.
- **The 51% Milestone:** The NBFIs sector now manages **\$256.8 trillion**, representing more than half (51%) of all global financial assets – *FSB 2025*.
- **Composition:** Investment funds (mutual, hedge), pension funds, and insurance companies play a dominant role.

## The "New Normal" for Financial Stability

Financial stability no longer depends solely on the solvency of banks, but on **market liquidity** and the resilience of investment funds.

# Structural Drivers of the NBFI Expansion

Why has the NBFI sector grown so aggressively?

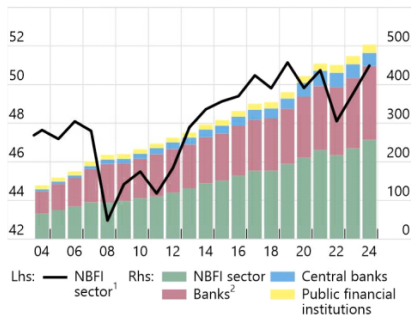
- ① **Regulatory Tightening (Push Factor):** Strict capital requirements on banks pushed riskier activities (e.g., mid-market lending) off bank balance sheets.
- ② **Search for Yield (Pull Factor):** The long period of low interest rates forced investors to seek returns in illiquid and alternative assets (Private Credit, Private Equity).
- ③ **Technology & Institutionalization:** Digitalization enabled lightning-fast capital moves and the rise of giant asset managers (BlackRock, Vanguard) with market power exceeding many central banks.
- ④ **Demographics:** Aging populations in developed countries increase assets in pension funds and life insurance.

# NBFI Assets as % of Total (FSB 2025)

Total global financial assets

Share of total financial assets

USD trillion



Composition of the global financial system<sup>4</sup>

|                                                   | Total global financial assets | Central banks | Banks <sup>2</sup> | PFIs <sup>3</sup> | NBFI sector |
|---------------------------------------------------|-------------------------------|---------------|--------------------|-------------------|-------------|
| <b>Size at end-2024 (USD trillion)</b>            | 503.7                         | 34.1          | 191.3              | 21.5              | 256.8       |
| <b>Share of total global financial assets (%)</b> | 100.0                         | 6.8           | 38.0               | 4.3               | 51.0        |
| <b>Growth in 2024 (year-over-year, %)</b>         | 6.3                           | -4.2          | 4.7                | 3.1               | 9.4         |
| <b>Growth 2018–23 (annualised growth, %)</b>      | 6.1                           | 6.6           | 6.3                | 5.2               | 5.9         |

<sup>1</sup> NBFI includes ICs, PFs, OFIs, and financial auxiliaries- <sup>2</sup> All deposit-taking corporations- <sup>3</sup> Public financial institutions.

Source: Jurisdictions' 2025 submissions (national sector balance sheet and other data) and 2021 submission for Russia; FSB calculations.

# NBFI Assets as % of Total by Country (FSB 2025)

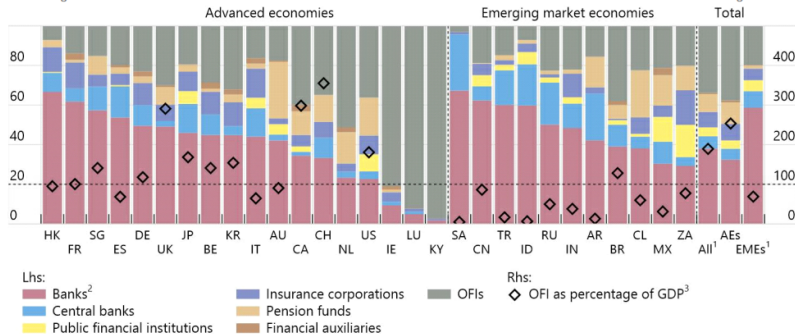
The structure of the financial system differed across jurisdictions, with banks comprising the single largest entity type in most jurisdictions

29-Group at end-2024

Graph 1-2

Percentage of total financial assets of domestic entities

Percentage of GDP



<sup>1</sup> RU not included in aggregates. <sup>2</sup> All deposit-taking corporations. <sup>3</sup> Jurisdictions with OFI assets greater (lower) than their GDP will be above (below) the horizontal dashed line. The percentage of OFI assets to GDP for the Cayman Islands (287,381), Luxembourg (20,854), Ireland (1,480) and the Netherlands (544) are not shown since they are particularly high compared to the rest of the jurisdictions.

Source: Jurisdictions' 2025 submissions (national sector balance sheet and other data); FSB calculations.

# Anatomy of the NBFI Sector: The "Narrow Measure"

NBFI is not a monolith. Regulators (FSB) focus on the **Narrow Measure** (\$76.3 trillion), consisting of entities that may pose systemic risk:

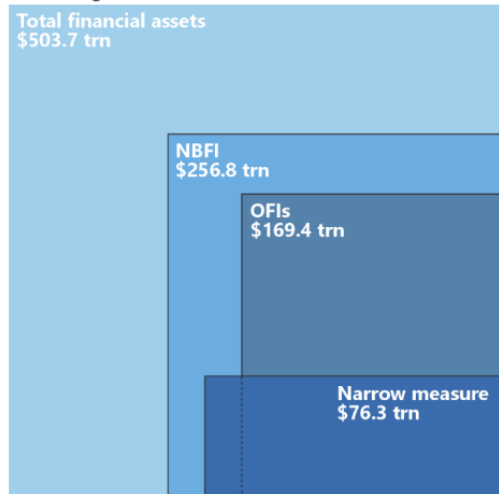
- **EF1: Investment Funds** (\$55T) – Vulnerable to liquidity mismatch (daily redemptions vs. illiquid assets).
- **EF2: Finance Companies** (\$4T) – Dependent on short-term wholesale funding, providing credit.
- **EF3: Broker-Dealers** (\$9T) – Key for market-making, highly interconnected through repo markets.
- **EF4: Insurance-led & Credit Enhancement** (\$0.5T) – Providing guarantees.
- **EF5: Securitization Vehicles** (\$7T) – Transforming risky loans into securities.

## Focus 2025–2026

The fastest-growing component within EF1 is now **Private Credit**, which is becoming a direct competitor to bank loans.

# The "Narrow Measure" (2025)

Narrowing down to the narrow measure<sup>1</sup>





# The Local Context: Investment Funds (CNB Perspective)

While global systemic risk often focuses on mega-funds, local markets are shaped by standard investment funds.

- **Retail vs. Qualified Investors:** Shift towards alternative investment funds (AIFs) alongside traditional UCITS.
- **Liquidity Transformation:** Funds offer daily liquidity to retail investors while holding less liquid assets (e.g., real estate, corporate bonds).
- **Systemic footprint:** Even without massive leverage, herding behavior among retail investors can cause local market disruptions.

## CNB's View on Vulnerabilities

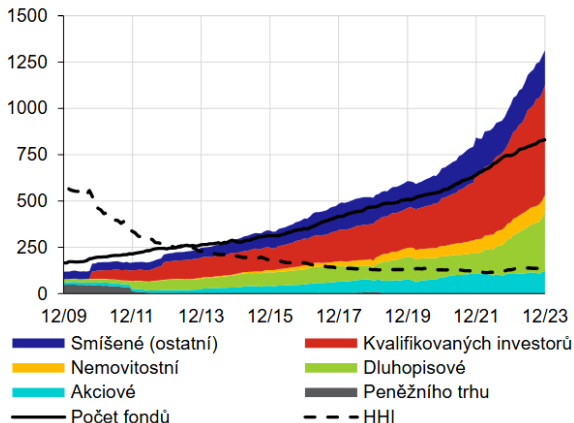
“Investment funds democratize capital markets, but their promise of daily liquidity can become a systemic vulnerability during market stress if underlying assets cannot be quickly liquidated.”

– *Adapted from CNB Blog (Everything you wanted to know about funds)*

# Investment Fund Assets in the Czech Republic

**Graf 6 – Aktiva segmentu českých investičních fondů**

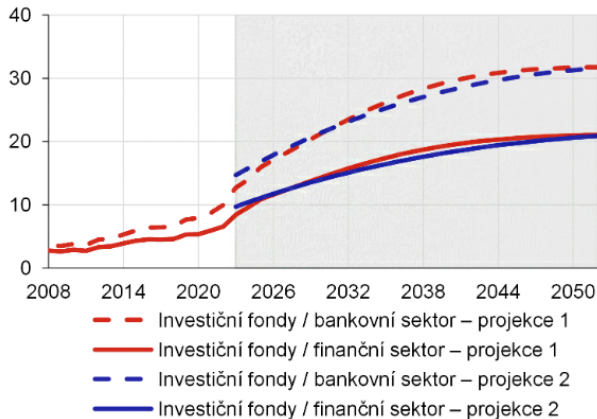
(v mld. Kč; počet fondů v jednotkách, HHI v rozmezí od 0 až 10 000)



Poznámka: Počet zahrnuje i podfondy. HHI značí Herfindahl – Hirschmanův index měření koncentrace. Fondy kolektivního investování jsou rozděleny dle převažujícího typu aktiv.

# Future Projections for the Czech Market

**Graf 14 – Projekce poměru aktiv investičních fondů k celkové výši aktiv finančního sektoru (v %)**



Poznámka: Šedá plocha označuje začátek projekce.

# The Wealth Shift: Czech Household Portfolios (CNB Blog)

- **The Migration:** Czech households are moving away from traditional overnight deposits toward **Investment Funds** and **Equities**.
- **Drivers:** Low real interest rates (historically) and a maturing capital market have pushed retail investors into the NBFIs sector.
- **The Risk Profile:**
  - **Market Risk Exposure:** Household wealth is now directly tied to stock and bond market volatility.
  - **Liquidity Expectation:** Retail investors often expect "instant" liquidity from funds that may hold illiquid underlying assets.

## The "Wealth Effect" Channel

A sharp correction in NBFIs asset prices now has a faster, more direct impact on **domestic consumption** (as households feel "poorer" immediately), creating a new macro-prudential concern for the CNB.

# The Hybrid Reality: Banks as Infrastructure Providers

The boundary between banks and NBFIs is blurring. Banks are no longer just "lenders" but have become **infrastructure providers** for NBFIs:

- **Prime Brokerage:** Lending securities and providing leverage (Hedge Funds).
- **Custody & Settlement:** Holding assets and clearing trades.
- **Liquidity Backstops:** Providing RCFs (Revolving Credit Facilities).

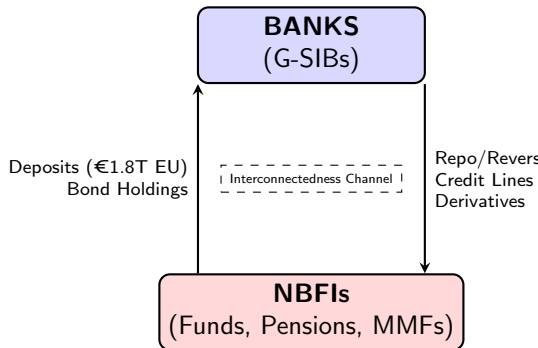
## Systemic Implication

Banks are now **indirectly exposed** to risks they cannot fully control (e.g., private equity strategies or highly leveraged crypto-proxies).

# Bank–NBFI Linkages

- **Banks fund NBFIs:** via repo, reverse repo, and committed credit lines (€1.66T in EU).
- **NBFIs fund banks:** via deposits and bank bond holdings (€1.8T in EU).
- **Off-balance-sheet ties:** Derivatives and securitisation pipelines.
- **High Concentration:** Euro-area G-SIBs intermediate 50–80% of derivatives/repos.

*Sources: FSB 2025, ECB 2026, IMF GFSR Oct 2025*



# The Core Nexus: Direct Funding (Balance Sheet)

## NBFIs Funding Banks (Liabilities Side)

- NBFIs account for roughly **15% of all bank funding** globally.
- In the Euro Area, NBFIs deposits in banks amount to **€1.8 trillion** (approx. 12.9% of all deposits) – *ECB 2026*.
- **Risk:** This wholesale funding is highly runnable. MMFs and investment funds will rapidly withdraw deposits during stress (as seen in March 2020).

## Banks Funding NBFIs (Asset Side)

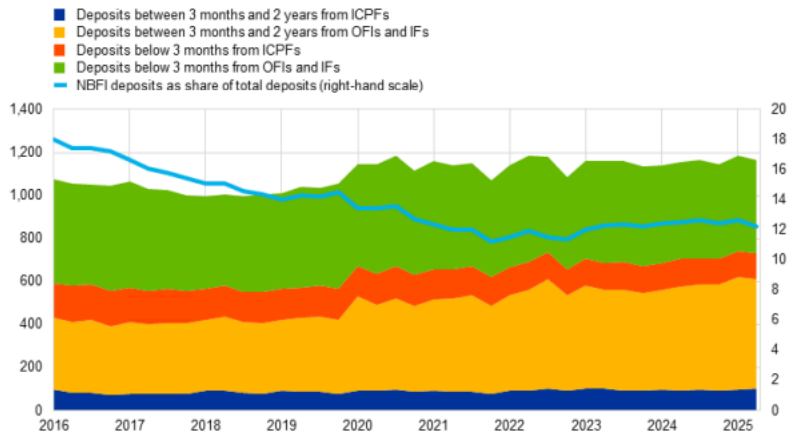
- Banks provide **€1.66 trillion** in direct credit to NBFIs in the EU (10% of total bank assets).
- This includes term loans, revolving credit facilities, and prime brokerage financing for hedge funds and private credit vehicles.
- **Risk:** Credit risk materialization if leveraged NBFIs default.

# NBFI Deposits in the EA Banking Sector

Share of deposits at euro area banks from NBFI entities, by maturity and business model

## a) Share of deposits at euro area banks from NBFI entities, by sector and maturity

(Q1 2016-Q2 2025; left-hand scale: EUR billions, right-hand scale: percentages)

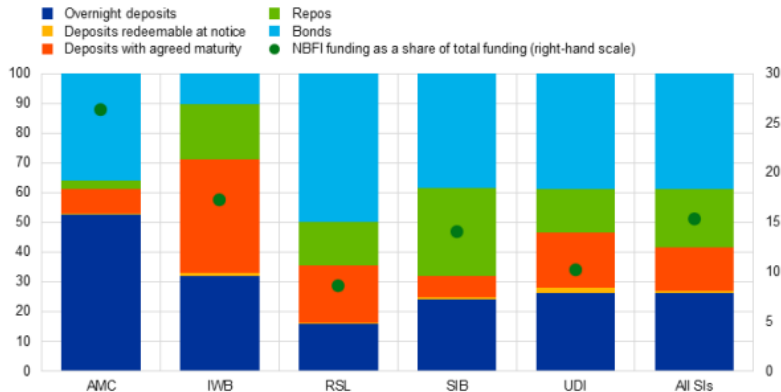




# NBFI Deposits by Bank Category

## a) Euro area banks' funding from NBFI entities

(Q2 2025, percentages)



*Highlights the concentration of NBFI funding within G-SIBs and large universal banks.*

# Transmission Channels: ECB 2026 Risk Matrix

The "Tightened Loop" means shocks in the NBFI sector transmit instantly to bank balance sheets.

| Risk Factor    | NBFI Vulnerability            | Bank Impact                 |
|----------------|-------------------------------|-----------------------------|
| Rates ↑        | Valuation losses (Bonds)      | Margin calls on derivatives |
| CRE Downturn   | Liquidity mismatch (RE Funds) | Credit losses on NBFI loans |
| Credit Spreads | Forced selling (Fire sales)   | MTM losses on trading books |

- **The Liquidity Trap:** Margin procyclicality forces NBFIs to sell assets, which lowers collateral values for banks, creating a feedback loop.

# Off-Balance Sheet: The Hidden Leverage

- **Repo Markets:** Banks are net borrowers of cash from MMFs and net lenders of cash to hedge funds.
- Reverse repos to hedge funds have **doubled since 2021** (ECB 2026), increasing exposure to highly leveraged players.
- **Derivatives:** NBFIs use banks to access interest rate and FX swaps (notional values exceed \$90 trillion globally).
- **Margin Procyclicality:** During market shocks, banks (as clearing members) must issue massive margin calls to NBFIs, forcing NBFI asset fire-sales.

## The Archegos Scenario

Synthetic leverage via Total Return Swaps (TRS) allows NBFIs to build massive exposures off-balance sheet. When positions move against them, prime broker banks absorb the ultimate losses.

# Indirect Linkages: Portfolio Overlap Channel

Exposures exist even **without direct lending** through common asset holdings.

- **Cosine Similarity Metric:** Measures portfolio overlap. Similarity between Euro-area banks and Pension Funds is  $> 0.90$  for sovereign bonds.
- **Fire-Sale Contagion Loop:**
  - ① NBFI faces investor redemptions or margin calls.
  - ② NBFI is forced to liquidate highly rated assets (e.g., government bonds, blue-chip corporate debt).
  - ③ Asset prices plummet (yields spike).
  - ④ **Impact on Banks:** Mark-to-market losses on bank trading books, depletion of CET1 capital, and reduced collateral values for bank borrowing.

# Step-In Risk: Banks as "Lenders of Last Resort"

Banks often act as the ultimate "lenders of last resort" for the NBFIs sector, creating contingent liabilities that only materialize during crises.

## Credit Lines & Liquidity Facilities

- Banks provide massive uncommitted and committed credit lines to Private Equity and Private Credit funds.
- In a crisis, funds draw down these lines simultaneously, draining bank liquidity exactly when the bank needs it most.

## Reputational Step-In

- Banks sponsor off-balance sheet vehicles (e.g., securitisation SPVs, MMFs).
- Even without a legal obligation, banks may "step in" to rescue a failing affiliated fund to protect their franchise reputation, bringing hidden toxic assets back onto the bank's balance sheet.

## Systemic Outlook: Concentration & Stress

- **Extreme Concentration:** The interconnectedness is not evenly distributed. A handful of Global Systemically Important Banks (G-SIBs) intermediate 50–80% of all NBFIs derivative and repo transactions.
- **Substitutability Risk:** If one major prime broker steps back, heavily leveraged NBFIs may collapse due to a lack of alternative funding.

### IMF 2025 Stress Test Findings

- In a severe NBFIs stress scenario, bank CET1 ratios could deplete by an additional **100 basis points** purely due to NBFIs contagion.
- **Liquidity drain:** Up to 14% of banks would face negative net liquidity if NBFIs simultaneously withdraw their deposits and max out their credit lines.

# Private Credit: The "Shadow" Lending Engine

Private credit has evolved from a niche asset class to a **\$2.1 trillion** global market, competing directly with traditional syndicated loans.

- **The Core Mechanism:** Non-bank lending to corporate borrowers (mid-market) via privately negotiated, non-traded loans.
- **Regulatory Arbitrage:** Stringent Basel III/IV requirements made mid-market lending "capital-expensive" for banks, shifting risk to the NBFI sector.
- **Borrower Appeal:** Speed, confidentiality, and "bespoke" terms compared to public bond markets.
- **Market Dominance:** In the US, private credit now competes directly with the syndicated loan and high-yield bond markets.

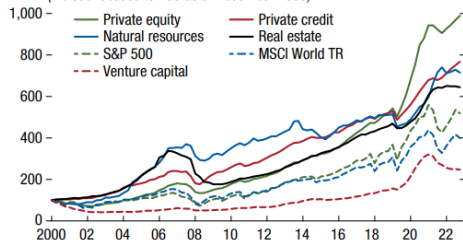
## The Shift in Risk

Credit risk hasn't disappeared; it has migrated from regulated bank balance sheets to "darker" corners of the financial system with lower transparency.

# The Exponential Growth of Private Credit

Private credit funds have delivered comparatively higher gross returns ...

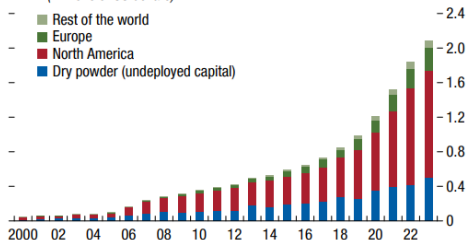
**1. Returns of Private Equity, Private Credit, and Other Asset Classes**  
(Indices rebased to 100 as of December 2000)



Private credit fund managers based in North America manage a material part of the market in other regions.

... and grown exponentially over the past two decades.

**2. The Growth of Private Credit Markets**  
(Trillions of US dollars)



In the United States, private credit size is comparable to leveraged loans and high-yield bond markets.

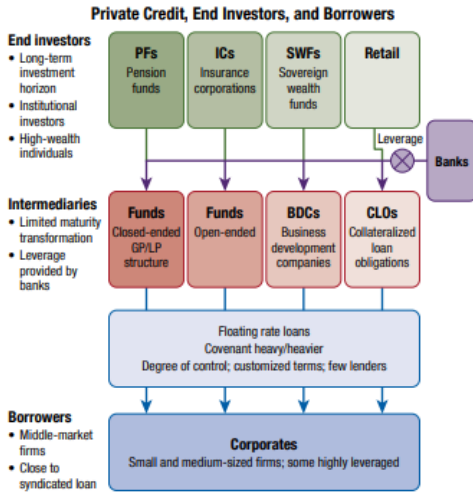
*Global AUM has seen a tenfold increase since the GFC, dominated by US-centric funds (87%) but funded by global institutional capital.*



# Private Credit Fund Structure

**Figure 2.1. Private Credit Structure**

Private credit funds are intermediaries between end investors and corporate borrowers that offer floating rate loans to middle-market firms.



# The Symbiosis: Private Credit and Private Equity (PE)

Private credit and Private Equity are two sides of the same coin. According to **IMF GFSR 2024**, over **70% of private credit deals** are "sponsored" by PE firms.

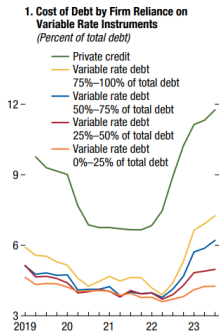
- **The Leveraged Buyout (LBO) Engine:** PE firms use private credit to finance the acquisition of companies. The private credit fund provides the debt, while the PE fund provides the equity.
- **Dry Powder Linkage:** As PE firms raise more capital ("dry powder"), the demand for private credit to leverage those investments grows proportionally.
- **"Extend and Pretend":** In times of stress, PE sponsors and private credit lenders can renegotiate terms privately (e.g., PIK - Payment-in-Kind interest), hiding defaults from the public eye.

## Inter-sectoral Contagion

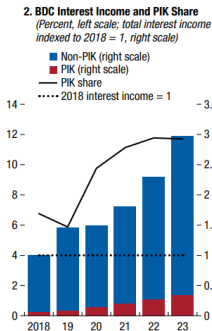
If a large PE sponsor faces a liquidity crisis, it affects not just its equity investors but dozens of private credit funds and the banks providing leverage to those funds.

# Structural Vulnerabilities: PIK and Covenants

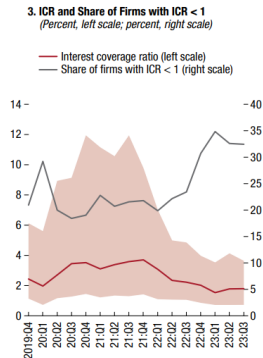
The transmission of higher rates into firms' cost of debt has been more swift for firms with variable rate debt.



Payment-in-kind interest payments have surged for BDC portfolios.



Public firms with size and leverage characteristics similar to private credit firm have shown a deterioration in their ability to pay interest.



Sources: BDC 10-K and 10-Q filings; S&P Capital IQ; and IMF staff calculations.

Note: Panel 1 shows the cost of debt, calculated as interest expense divided by total debt. Medians are taken for each bucket of variable rate debt reliance, whereby this reliance is expressed as the ratio of variable rate debt over total debt. The cost of debt within each bucket varies based on credit fundamentals. Private credit rates are based on a sample of BDC portfolios. In panel 2, when interest is paid in kind, no cash flow occurs. Instead, the interest coupon is added—usually at an extra cost—to the loan's principal. Statistics in panel 3 are based on a sample of public firms located in North America with size and leverage characteristics similar to those of borrowers in the private credit universe. It should be noted that interest coverage and debt/EBITDA ratios are usually not reflected in firm-level databases when earnings are negative. This means that the true number of firms with unsustainable interest expense level is (even) higher than indicated by the ICR = 1 threshold. BDC = business development company; EBITDA = earnings before interest, taxes, depreciation, and amortization; ICR = interest coverage ratio; PIK = payment in kind.

# Valuation Opacity and "Volatility Smoothing"

Private markets report lower volatility than public markets—but this is often an illusion created by accounting methods.

- **Mark-to-Model:** Pricing based on internal assumptions rather than market trades.
- **Infrequent Appraisals:** Lags in reflecting the true impact of high interest rates.

## Risk Implications

- Underpriced leverage.
- Delayed loss recognition.
- **Sudden "Cliffs":** Risk of sharp, lumpy devaluations when models finally catch up to reality.

# The Leverage Trap: "Leverage on Leverage"

The systemic risk is amplified by layers of debt within the private credit ecosystem.

- **Asset-Level:** Borrowers are already highly indebted (leveraged mid-market firms).
- **Fund-Level:** Private credit funds use **Subscription Lines** (bridge loans from banks) to boost internal rates of return (IRR).

## The Macro Concern (IMF 2025)

The sector grew during a "Goldilocks" period of low rates. It remains **untested** in a prolonged high-rate environment with a significant corporate default cycle.

# Transmission to the Banking Sector

How does a Private Credit "default wave" hit the regulated banks?

- **Direct Exposure:** Banks provide "subscription lines" (short-term bridge loans) to private credit funds.
- **Asset-Based Finance (ABF):** Banks are increasingly selling their own loan portfolios to private credit funds to clean up balance sheets, creating a feedback loop.
- **Wealth Management:** G-SIBs distribute private credit products to their high-net-worth clients. A collapse in valuations could lead to massive reputational damage and legal liabilities.

*"The lack of transparency and high interlinkage with systemic banks make private credit a potential 'blind spot' for financial stability."*

*— BIS Quarterly Review, 2025*

# Key Channels of Shock Transmission

## 1. Liquidity Mismatch

Daily redemptions vs. illiquid underlying assets (Real Estate, Private Credit).

## 2. Margin & Leverage

Procyclical margin calls and hidden synthetic leverage (Archegos).

## 3. Interconnected USD Funding

High reliance on FX swaps and global USD liquidity.

## 4. Correlation & Herding

High portfolio overlap ( $> 0.9$ ) leads to simultaneous "dashes for cash."

## The Shift

Algorithmic trading and digital connectivity mean shocks no longer travel linearly—they propagate exponentially through automated margin processes.

## Channel 1: Liquidity Spirals & Fire Sales

This channel transforms an individual fund's liquidity issue into a systemic price collapse.

- **The Trigger:** Sudden redemptions or margin calls in open-ended funds.
- **First-Mover Advantage:** Investors exit early to avoid being stuck with illiquid assets, creating a "run" on the fund.
- **The Forced Sale:** Funds sell their most liquid assets (Sovereign bonds) first, depressing prices for all holders (Banks, Pensions).
- **ECB 2026 Observation:** Increased **algorithmic trading** has made these spirals faster and more violent.

### March 2020: "Dash for Cash"

Even the US Treasury market—the world's most liquid—faced a shock as NBFIs sold everything to secure cash, requiring massive central bank intervention.

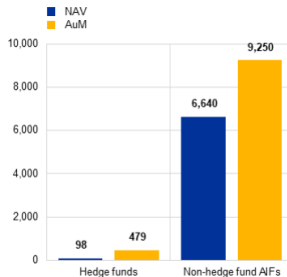


# EA Hedge Funds: Liquidity and Leverage

Hedge funds subject to AIFMD rules: net asset value and regulatory assets under management (panel a) and liquidity profile (panel b)

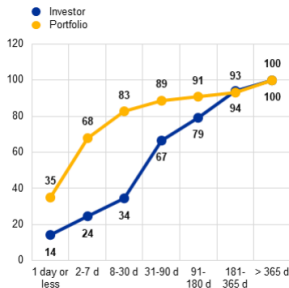
a) Net asset value and regulatory assets under management

(EUR billions)



b) Liquidity profile

(percentages)



Sources: ECB (AIFMD), NCAs and ESMA.

Notes: Data as at year-end 2024. Panel a): NAV and AuM by AIF type. "Regulatory assets under management" refers to the value of all portfolio assets, including all assets acquired through the use of leverage (borrowing of cash or securities and leverage embedded in derivative positions). Data refer to AIFs of authorised and registered AIFMs excluding feeder funds. Panel b): portfolio and investor liquidity profiles of hedge funds managed and/or marketed by authorised EEA AIFMs. Portfolio profile determined by percentage of the portfolios capable of being liquidated within each specified period; investor profiles depend on the shortest period within which redemption payments could be received. Data refer to AIFs of authorised managers excluding feeder funds and AIFs not reporting liquidity profile data.

*Note the widening gap between fund liquidity and investor redemption rights.*

## Channel 2: Procyclical Margining & Leverage Stack

- **Procyclicality:** Banks/CCPs increase margin requirements (IM/VM) exactly when volatility spikes, draining NBFI cash precisely when it is most scarce.
- **The Leverage Stack:** Leverage is now multi-layered:
  - ① **Asset-Level:** Highly indebted corporate borrowers (LBOs).
  - ② **Fund-Level:** Subscription lines and repo borrowing.
  - ③ **Synthetic:** Off-balance sheet Total Return Swaps (TRS).

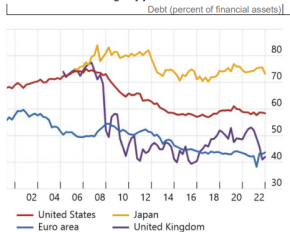
### Hidden Risk: The Archegos Case

Synthetic leverage via swaps allowed one entity to build massive exposures across multiple banks; none of the banks saw the "total" risk until the margin call.

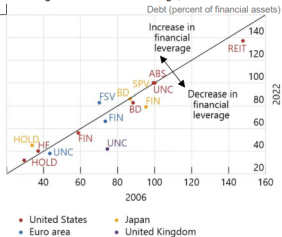
# Financial vs. Synthetic Leverage Trends

## Direct Financial Leverage

3. OFI financial leverage by jurisdiction



4. Change in OFI financial leverage, 2006-22<sup>4</sup>



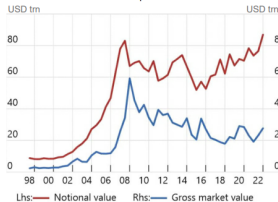
ABS = Issuers of asset-backed securities; BD = broker-dealers; FIN = finance companies; FSV = finance securitisation vehicles; HF = hedge funds; HOLD = holding companies; REIT = real-estate investment trusts; SPV = special purpose vehicles; and UNC = uncategorised OFIs.

## Synthetic Leverage (Derivatives)

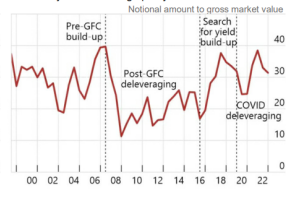
NBFI OTC derivatives and synthetic leverage<sup>1</sup>

Graph 5

1. NBFI OTC derivatives exposures



2. NBFI synthetic leverage proxy



<sup>1</sup> The charts cover FX options as well as interest rate, equity and credit derivatives for non-bank entities. CCPs are excluded from the charts. Sources: BIS; FSB calculations.

## Channel 3: Cross-Border Spillovers & USD Funding

The global NBFIs sector depends heavily on U.S. Dollar funding, regardless of its home base.

- **The FX Swap Nexus:** Non-US funds use short-term FX swaps to hedge USD assets. Off-balance sheet USD obligations have reached **\$80 trillion** (BIS 2025).
- **Refinancing Risk:** In the "Higher-for-Longer" regime, duration-heavy portfolios face constant valuation pressure and refinancing "cliffs."
- **Global Contagion:** A US Hedge Fund failure directly hits its European G-SIB Prime Broker, restricting credit to local EU firms.

## Channel 4: The Correlation Channel (Herding)

Market intermediation and portfolio choices are now extremely concentrated.

- **Herding:** High use of ETFs and algorithmic strategies means diverse players act as a single block during stress.
- **Plumbing Risk:** A small number of G-SIBs dominate Repo, Prime Brokerage, and Clearing.
- **Correlation** → **1:** In a crisis, the "safe" and "risky" assets start moving together, breaking traditional diversification models.

*"If the central 'plumbing' (G-SIBs) retreats from market-making due to their own risk limits, liquidity vanishes for the entire system."*

# Synthesis: The Multiplier Effect

## The Chain Reaction:

- ① **Shock:** Interest rates rise or CRE defaults spike.
- ② **Valuation:** Bond prices fall → Margin calls issued.
- ③ **Liquidity:** NBFIs dump blue-chips to meet calls.
- ④ **Contagion:** Banks face losses on credit lines and de-risk.
- ⑤ **Impact:** Credit crunch for the real economy.

**Conclusion:** NBFIs risk is no longer "contained" outside the banking system; it is the primary driver of modern financial instability.

## Case Study 1: March 2020 – The "Dash for Cash"

- **The Shock:** Pandemic uncertainty triggers a global flight to safety.
- **The Paradox:** Even US Treasuries (the ultimate safe asset) saw prices plummet and yields spike due to unprecedented NBFIs selling pressure.
- **Transmission:**
  - **Open-ended funds** faced massive redemptions and sold liquid assets first.
  - **Hedge funds** unwinding "basis trades" (arbitrage between cash and futures) accelerated the fire-sales.
  - **Bank dealers** were constrained by post-GFC balance sheet regulations (SLR), preventing them from absorbing the selling volume.
- **Lesson:** In a systemic crisis, NBFIs liquidity needs can easily overwhelm the banking sector's capacity to act as a market maker.

# Case Study 1: March 2020 – Yields

Cumulative Yield Changes across Sovereign Bond Markets



Source: Bloomberg L.P.

Notes: The chart displays the cumulative yield changes for ten-year sovereign bonds, starting on January 1, 2020. U.S., Germany, U.K., and Japan denote Treasury, bund, gilt, and Japanese government bond (JGB) securities, respectively.

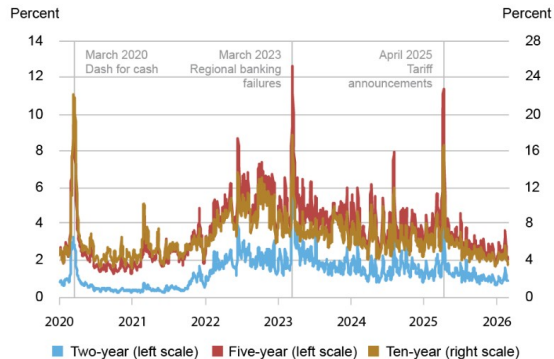
Source:

<https://libertystreeteconomics.newyorkfed.org/2022/07/the-global-dash-for-cash-in-march-2020/>



# Case Study 1: March 2020 – Yields

Volatility Trended Down after April 2025 Spike



Source:

<https://libertystreeteconomics.newyorkfed.org/2026/04/treasury-market-liquidity-since-april-2025/>

## Case Study 2: Archegos (2021) – Synthetic Leverage Risks

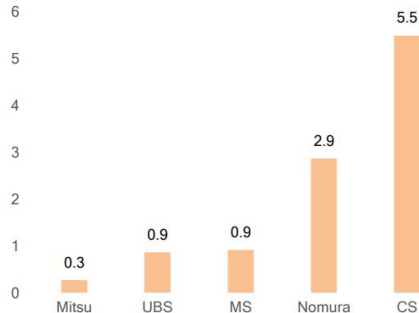
- **Mechanism:** A family office used **Total Return Swaps** (TRS) with multiple global G-SIBs to build a concentrated \$50B+ exposure.
- **The Failure:** Because the leverage was synthetic (off-balance sheet), no single prime broker knew the true size of the client's total position.
- **Impact:** A margin call triggered a fire-sale, resulting in **\$10B+** in losses for banks like Credit Suisse and Nomura.

### Regulatory Lesson

Transparency in derivatives is a "blind spot." Prime brokers must assess **aggregated leverage** across the entire financial system, not just their own books.

## Case Study 2: Archegos (2021) – Direct Losses

Direct losses related to Archegos  
More than USD 10bn in losses

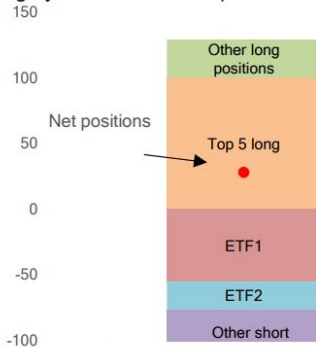


Note: Reported losses in USD bn as of April 2021. Mitsu: Mitsubishi UFG; MS: Morgan Stanley; CS: Credit Suisse.  
Sources: Company disclosures, ESMA.

## Case Study 2: Archegos (2021) – Concentration Risk

Archegos' exposures as of 26 February 2021

Highly concentrated exposures



Note: Long and short positions of Archegos by underlying, rebased as of 26 Feb. 2021. Data for EEA30 counterparties.

Sources: EMIR, ESMA.

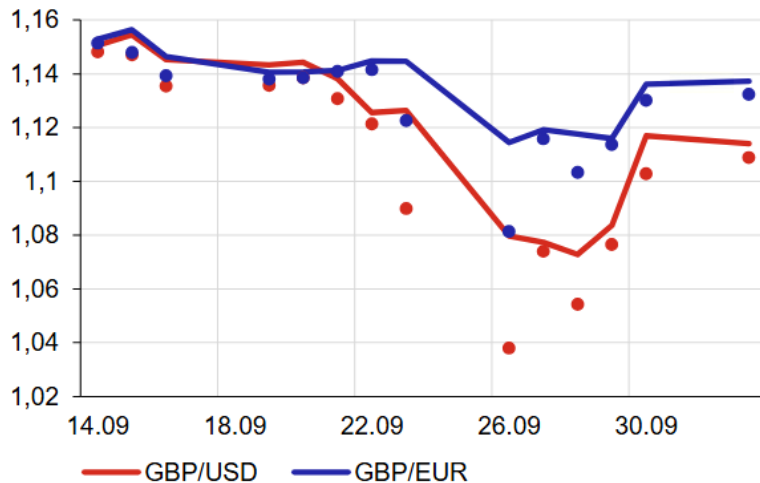
## Case Study 3: UK LDI Crisis (2022) – The Pension Fund Trap

- **Context:** UK pension funds used Liability-Driven Investment (LDI) strategies with **3x leverage** to match future liabilities.
- **Shock:** Rapid rise in Gilt yields (triggered by the "mini-budget") led to massive margin calls on LDI derivative positions.
- **The Death Spiral:** To raise cash for margin, funds sold Gilts → Gilt prices fell → yields rose → more margin calls.
- **Bank Intervention:** The Bank of England had to act as "Buyer of Last Resort" to prevent a systemic collapse of the pension sector.
- **Lesson:** Even "conservative" NBFIs (pension funds) can become systemic risks through leveraged hedging strategies.

# LDI Crisis: Impact on Gilt Prices

Graf 2 – Kurz GBP/EUR a GBP/USD

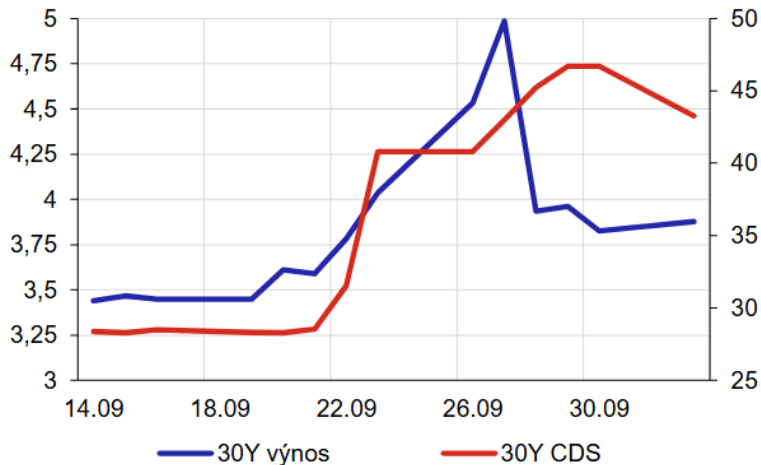
(jednotky měny za 1 GBP)



# LDI Crisis: Impact on Gilt Yields

**Graf 3 – 30letý výnos dluhopisů a CDS sazby v UK**

(v %; pravá osa: v bazických bodech)



## Case Study 4: SVB (2023) – The NBFIs Deposit Connection

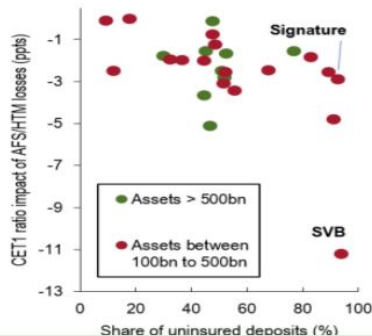
- **The Linkage:** Silicon Valley Bank (SVB) had a deposit base highly concentrated in **VC and Private Equity funds**.
- **Transmission:** As VC funding dried up, these NBFIs clients withdrew deposits simultaneously.
- **The Digital Run:** Social media and digital banking accelerated the flight, with \$42B in withdrawal requests in 24 hours.
- **Lesson:** Sectoral concentration in bank funding (relying on NBFIs) creates extreme vulnerability. **Correlation of depositor behavior** is a modern systemic risk.



## Case Study 4: SVB (2023) – HTM losses

### *Unrealized losses and deposit mix*

**2. Share of Uninsured Deposits and Impact on Available-for-Sale (AFS)/Held-to-Maturity (HTM) Losses on CET1 Ratio as of fourth quarter 2022 (Percentage)**

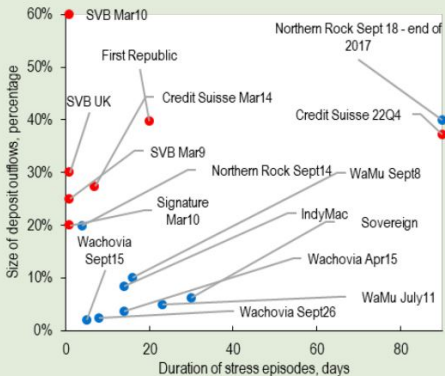


## Case Study 4: SVB (2023) – Bank Runs

## Fastest and largest deposit run

### 3. Deposit Runs

(Percentage of total deposits and number of days)



## Comparative Analysis: What have we learned?

| Episode    | Core Driver           | Key NBFI Sector    | Systemic Impact                   |
|------------|-----------------------|--------------------|-----------------------------------|
| March 2020 | Market Liquidity      | MMFs / Hedge Funds | Global Central Bank Intervention  |
| Archegos   | Hidden Leverage       | Family Offices     | Severe Bank Capital Losses        |
| UK LDI     | Margin Spirals        | Pension Funds      | Sovereign Bond Market Instability |
| SVB 2023   | Funding Concentration | VC / PE Funds      | Regional Banking Crisis           |

### Key Takeaways for Students

- **Leverage is often hidden** until a price shock occurs.
- **Liquidity mismatch** is the primary "fuel" for systemic fire-sales.
- **Interconnectedness** means that a problem in a "small" fund can threaten G-SIBs.

# Geopolitical Fragmentation as a Financial Risk

- **Capital Misallocation:** Cross-border flows are increasingly driven by geopolitical alignment ("friend-shoring") rather than fundamentals.
- **Shock Amplification:** Geopolitical events trigger rapid capital flight from emerging markets, straining local bank liquidity and NBFIs.
- **Inflationary Pressure:** Reshoring supply chains leads to "higher-for-longer" interest rates, putting permanent pressure on leveraged NBFIs.

# Policy Implications: Strengthening the Perimeter

## 1. Enhanced Macroprudential Monitoring

Mandate granular data reporting on non-bank leverage and improve cross-border sharing via global Trade Repositories.

## 2. Targeted NBFIs Resilience

Shift from bank-centric rules to NBFIs frameworks:

- Liquidity buffers and redemption "gates" for open-ended funds.
- Robust valuation standards for Private Credit (Mark-to-Market requirements).

## 3. Structural Market Reforms

Expand **Central Clearing** for government bonds and repos to ensure "market-making" doesn't freeze during stress.

# A Taxonomy of Modern Financial Fragility

**Systemic risk today is layered, not localized.**

- **1. Micro-Level (Entity):** Liquidity mismatches, hidden leverage, and valuation opacity in individual funds.
- **2. Meso-Level (Network):** High concentration in prime brokerage and recursive collateral chains.
- **3. Macro-Level (Regime):** Higher-for-longer rates and USD funding dependencies.

## Central Insight

The system is safer at the core (well-capitalized banks), but significantly more fragile at the edges (the NBFI "blind spots").

## Conclusion: The New Financial Landscape

- **Unprecedented Integration:** NBFIs now hold **51% of global financial assets**. The system is no longer bank-centric.
- **Risk Migration:** Risk has not been eliminated; it has moved to less transparent sectors (Private Credit, Hedge Funds).
- **Persistent Linkages:** The banking sector (G-SIBs) is highly capitalized, but their deep linkages to NBFIs mean they remain exposed to off-balance-sheet contagion.
- **Future Tests:** Geopolitical fragmentation and structurally higher interest rates will rigorously test the resilience of this deeply integrated, highly leveraged ecosystem.

**The system is more resilient in parts, but more interconnected everywhere.**

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